

MIKHAIL EMAD ALBER

Software Engineer | Full-Stack & Systems Architecture

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PROFESSIONAL SUMMARY

Results-oriented Software Engineer with extensive experience architecting and deploying scalable web services, high-throughput REST APIs, and automated data pipelines. Delivered 40+ engineered applications leveraging Python (Django, Flask), JavaScript (React, Node.js), relational & NoSQL databases (PostgreSQL, SQL Server, MongoDB), and Linux server infrastructure. Proven expertise in microservices architecture, JWT/RBAC security, real-time WebSockets, database query optimization, and production environments maintaining high availability.

WORK EXPERIENCE

Systems & Software Support Engineer

Petra Software | 2025 - Present | Cairo, Egypt

- Administer and maintain production Ubuntu Linux server environments across SiteGround and cPanel, sustaining high application uptime and reliability for enterprise client portals.
- Deploy and configure full-stack web applications on live cloud instances, implementing automated verification checks, SSL/DNS configurations, and continuous release rollouts.
- Isolate and resolve tier-2/3 system defects, database deadlocks, and API bottlenecks, coordinating with core development teams to accelerate issue resolution.
- Optimize SQL database configurations, query performance, and server resource utilization, establishing standard operating procedures and technical troubleshooting runbooks.

TECHNICAL PROJECTS

Enterprise Restaurant POS & Operations Platform

React | Django REST | PostgreSQL | WebSockets | JWT

- Engineered high-concurrency order dispatch utilizing real-time WebSockets, decreasing order transmission latency to under 250ms across 50+ concurrent POS terminals.
- Automated multi-station kitchen display routing and network thermal printing across 3 prep stations, reducing order fulfillment transit times by 45%.

Hospital Clinical & Operations Management System

React | Django REST | PostgreSQL | RBAC Security

- Architected enterprise healthcare suite coordinating electronic health records (EHR), outpatient scheduling, and financial ledgers across 8 hospital departments.
- Automated real-time pharmacy inventory deduction upon prescription issuance, eliminating dispensing errors and accelerating cross-department billing audits by 50%.

Retail Data Warehouse & Analytics Platform

Python | SQL Server | MongoDB | Flask | Pandas | ML

- Constructed an end-to-end ETL data pipeline ingesting 100,000+ relational transactions from SQL Server into MongoDB, exposing aggregated analytics via sub-100ms Flask APIs.
- Integrated predictive machine learning algorithms for sales analysis and inventory demand forecasting, reducing stockout risks by 28% and optimizing supply chain reorders.

Real-Time Edge Computer Vision & Detection Engine

Python | OpenCV | TensorFlow / Keras | MobileNet CNNs

- Built an edge-optimized computer vision inference pipeline simultaneously evaluating mask compliance and classifying multi-class emotional states on live video streams with 94%+ accuracy at 30+ FPS.

EDUCATION

Bachelor of Science in Computer Science & Engineering

The Egyptian E-Learning University (EELU)

Expected 2025

TECHNICAL SKILLS

BACKEND & APIS

Django Django REST Flask Node.js
Express RESTful APIs WebSockets
JWT Auth Microservices

FRONTEND & UI

React.js Next.js Redux
State Management Responsive Design
Bootstrap Tailwind CSS Three.js / WebGL

DATABASES & STORAGE

PostgreSQL Microsoft SQL Server MySQL
MongoDB Redis Schema Design
Indexing Query Optimization

DEVOPS & CLOUD

Linux / Ubuntu Server Docker Git
GitHub CI/CD Nginx cPanel
SiteGround Reverse Proxy

AI & DATA ENGINEERING

ETL Pipelines Data Analysis
Machine Learning Deep Learning OpenCV
TensorFlow Keras CNNs NLP
Transformers Pandas

LANGUAGES

Python JavaScript (ES6+) TypeScript
SQL HTML5 CSS3 Bash Shell

METHODOLOGIES & CORE

System Design OOP Data Structures
Agile / Scrum Root-Cause Analysis

CERTIFICATIONS

Data Science & AI Specialization

AMIT Learning

2026

Full-Stack Web Development

AMIT Learning

2024 - 2025

AI & Cloud Computing

Huawei Technologies

2022

LANGUAGES

English: Professional (B2)

Arabic: Native Proficiency